

Optimal Transport

Amp

How can we rearrange a given amount of material into a desired shape at the least cost? Four spoonfuls of sand lead to transport plans, conservation constraints, and distances between probability distributions. A visual construction on a ruler explains why cumulative distributions solve one-dimensional transport. An optional computational extension introduces entropy and Sinkhorn scaling. The starting prerequisites are arithmetic and coordinates on a line.

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Contents

1. Moving a shape, one piece at a time	3
2. A grid that keeps every piece accounted for	4
3. From a cheaper plan to a proven optimum	6
4. From sand to a distance between distributions	7
4.1. Why squared distance needs a square root	8
5. Solving transport on a ruler	9
5.1. Count each trip by the gates it crosses	9
5.2. Build such a plan by consuming piles in order	10
6. How a computer chooses the routes	12
6.1. Optional extension: why add entropy?	12
6.2. Derive the two scaling updates	13
6.3. What the approximation guarantees, and what to check	14
7. Sources and next steps	15
Bibliography	16

1. Moving a shape, one piece at a time

Imagine four equal spoonfuls of sand on a ruler: **three at position 0, one at position 3**. You want **two at position 1, two at position 4**. You may divide the sand; nothing may be created or lost. For now, moving one unit one ruler step costs one unit of effort. How can we reshape the piles with the least effort?

The important constraint is the **whole target shape**. Sending every piece to its closest destination would send all three units at 0 to position 1, which only needs two. We must coordinate the trips. The sand picture is also useful when the objects are probabilities or data rather than physical cargo [1].

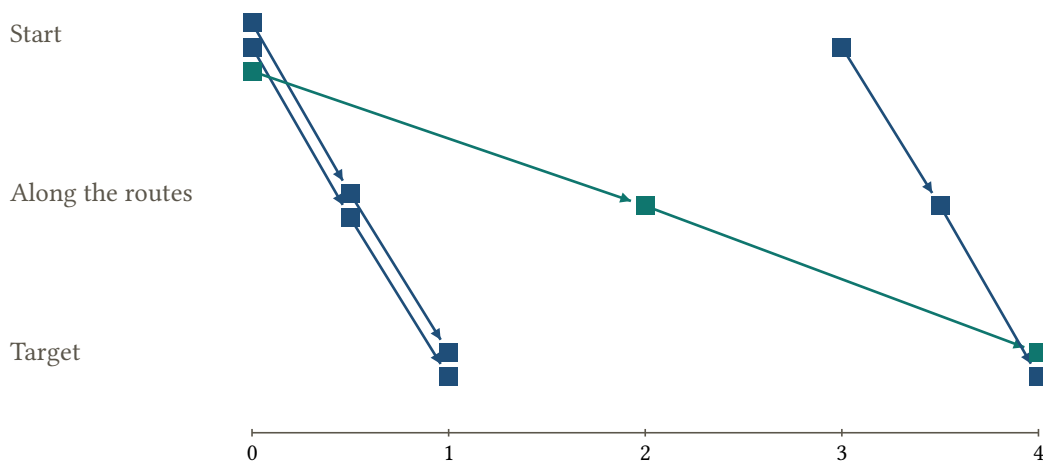


Figure 1: Follow the four squares from top to bottom. Each square is one unit of sand. Arrows show the chosen routes; the middle row illustrates halfway along those routes. The target has the same four units, rearranged.

Read the arrows as instructions: send **two** units from 0 to 1, **one** from 0 to 4, and **one** from 3 to 4. We call this list a **transport plan**. It meets both demands: 1 receives two, and 4 receives one plus one.

The middle row is just a way to follow the pieces. The basic transport problem specifies where they start, where they end, and the cost of each route. It does not specify travel times or velocities.

Cost is amount $\times$ distance, added over the routes. This plan costs

$$2 \times 1 + 1 \times 4 + 1 \times 1 = 7. \quad (1)$$

We have found a valid plan. To decide whether it is the cheapest one, we need a way to describe **all** valid plans without drawing every grain of sand.

2. A grid that keeps every piece accounted for

A matrix is a rectangular grid of numbers. Here it simply stores the route amounts we already understand:

$$P = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}. \quad (2)$$

The first row records sand leaving 0; the second records sand leaving 3. The first column records sand arriving at 1; the second records sand arriving at 4. Thus the upper-right entry, 1, is the teal square's route **from 0 to 4**.

ROUTE AMOUNTS	To 1	To 4	SENT
From 0	2	1	3
From 3	0	1	1
Received	2	2	4 total

Table 1: Sum horizontally to account for a starting pile; sum vertically to account for a destination. The four central cells are amounts sent along routes.

Row totals are the starting piles (3 and 1); column totals are the desired piles (2 and 2). The entries cannot be negative: a negative shipment is not an amount of sand. **The grid records movements, not distances.**

For a larger problem, label the sources $i = 1, \dots, n$ and destinations $j = 1, \dots, m$. These labels count places; they are not their coordinates. Let a_i be the amount initially at source i , b_j the amount wanted at destination j , and P_{ij} the amount sent between them.

The symbol $\sum$ means “add all these terms.” Summing across each row and down each column says precisely what the figure says:

$$P_{ij} \geq 0, \quad \sum_{j=1}^m P_{ij} = a_i, \quad \sum_{i=1}^n P_{ij} = b_j. \quad (3)$$

The row equation applies to every source; the column equation to every destination. Consequently $\sum_i a_i = \sum_j b_j$ must hold. This is **balanced transport**: all material is retained. Unequal totals require changing the model, for example by explicitly pricing creation or disposal.

Now store the **per-unit costs** in a separate grid, C . For our ruler, subtract coordinates and take the positive distance:

$$C = \begin{pmatrix} |0-1| & |0-4| \\ |3-1| & |3-4| \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 2 & 1 \end{pmatrix}. \quad (4)$$

Multiply corresponding entries of P and C , then add. This is the same amount $\times$ distance calculation as before. A feasible plan is one satisfying Equation 3. In general the optimization becomes

$$L_C(a, b) = \min_{P \text{ feasible}} \sum_{i=1}^n \sum_{j=1}^m P_{ij} C_{ij}. \quad (5)$$

Here min asks for the smallest possible value. In words: **choose the route amounts, keep every supply and demand fixed, and minimize the total cost**. This is the discrete **Kantorovich formulation** [1].

Why allow a source to send to several destinations? Our pile at 0 contains three units but the target at 1 only needs two. A rule that sends the entire pile at each source to a single destination cannot realize this target. Such a single-destination rule is called a **transport map**; the more flexible plan allows splitting. Treating individual grains as separate sources could also describe this example, but that uses a finer representation of the input.

3. From a cheaper plan to a proven optimum

To see what optimization changes, start with a different valid plan: send one unit from 0 to 1, two from 0 to 4, and one from 3 to 1. Its cost is $1 + 2 \times 4 + 2 = 11$. Keep one 0-to-1 trip and one 0-to-4 trip fixed. We will change only the **other two** units.

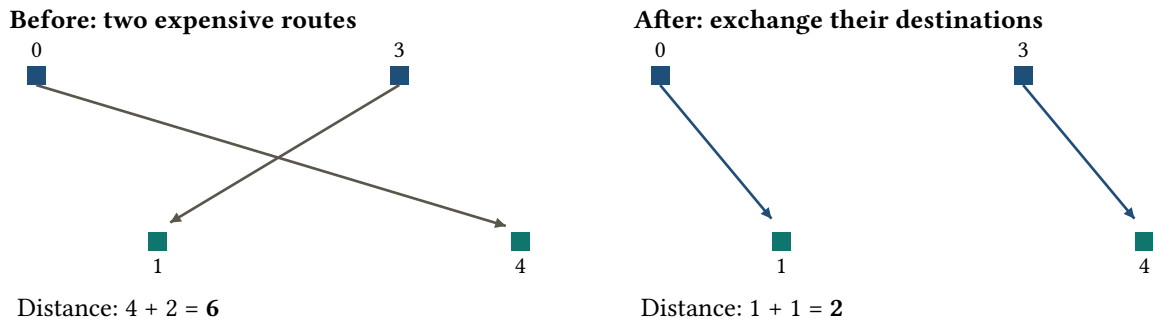


Figure 2: Each panel follows the same two units. Each source still sends one and each destination still receives one. Exchanging their destinations saves four ruler steps. Arrow direction means travel from the upper position to the lower position.

This swap preserves every pile total. Its only effect is to replace two costly trips with two short trips. The full plan's cost falls from 11 to 7:

$$\begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}. \quad (6)$$

Now we have a reason for the improvement, but we still need a reason no other plan beats 7. Add the positions of all four units. Initially the sum is $3 \times 0 + 1 \times 3 = 3$; finally it is $2 \times 1 + 2 \times 4 = 10$. So the sand must make **seven net steps to the right**.

Any leftward movement would add travel that must be compensated by extra rightward movement. Total distance therefore cannot be less than seven. Our plan moves every unit rightward and uses exactly seven steps. It attains that unavoidable lower bound, so **7 is the optimal cost**. This argument also allows splitting sand into arbitrarily small pieces: each piece contributes its amount multiplied by its displacement.

There is also a way to solve this example by bookkeeping alone. Call the upper-left entry t . The first row must sum to 3, so the upper-right entry is $3 - t$. The first column must sum to 2, so the lower-left entry is $2 - t$. Finally the second row must sum to 1, giving the last entry $t - 1$:

$$P(t) = \begin{pmatrix} t & 3-t \\ 2-t & t-1 \end{pmatrix}, \quad 1 \leq t \leq 2. \quad (7)$$

The bounds come from keeping all four entries nonnegative. Fractional values of t are allowed. Substitute these route amounts into the cost:

$$t + 4(3 - t) + 2(2 - t) + (t - 1) = 15 - 4t. \quad (8)$$

Increasing t by one lowers cost by four: exactly the exchange in the picture. Taking the largest allowed value, $t = 2$, gives the minimum cost 7.

4. From sand to a distance between distributions

For probability distributions, divide every amount here by four: the total mass becomes one. A **distribution** records where that one unit is located. It can describe probabilities: 3/4 at position 0 means a randomly selected piece starts there with probability 3/4. The data are now

$$a = \left(\frac{3}{4}, \frac{1}{4}\right), \quad b = \left(\frac{1}{2}, \frac{1}{2}\right), \quad P^* = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{4} \end{pmatrix}. \quad (9)$$

The coordinates remain $x = (0, 3)$ and $y = (1, 4)$. Weights alone do not say where the mass is. Two distributions with the same weights at different locations can be far apart.

You may see the compact notation

$$\mu = \frac{3}{4}\delta_0 + \frac{1}{4}\delta_3, \quad \nu = \frac{1}{2}\delta_1 + \frac{1}{2}\delta_4. \quad (10)$$

Read δ_x as “one unit concentrated at position x .” The letters μ and ν name the whole distributions, combining locations and weights. There is no need to treat a point mass as an ordinary smooth function.

After normalization, each plan entry is a fraction of the whole population. It is the probability of the paired event “starts at x_i and finishes at y_j .” A plan with these prescribed starting and ending probabilities is also called a **coupling**. Its average travel distance is

$$\sum_{i,j} P_{ij} |x_i - y_j|. \quad (11)$$

Minimizing that average gives the **1-Wasserstein distance**, written W_1 :

$$W_1(\mu, \nu) = \frac{1}{2} \times 1 + \frac{1}{4} \times 4 + \frac{1}{4} \times 1 = \frac{7}{4} = 1.75. \quad (12)$$

This is how a distance **between points** becomes a distance **between whole distributions** [1]. The material need not actually move; the cheapest hypothetical rearrangement measures how different the distributions are.

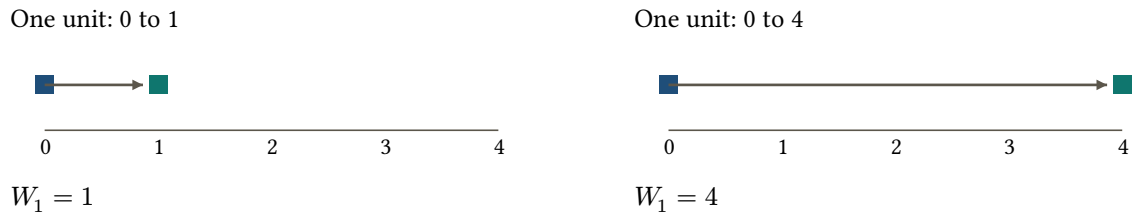


Figure 3: The same amount is moved in both cases. A location-aware comparison sees that the right-hand move is four times as far.

If we compared the two histograms bin by bin on positions 0,1,2,3,4, either move above would remove one from bin 0 and add one to a new bin. The sum of absolute bin differences would be 2 in either case. That comparison counts changed mass but has no notion of the distance between bins. Transport retains that geometric information through C .

Even the mean position is insufficient. Half the mass at -1 and half at 1 has mean 0 ; a single unit at 0 also has mean 0 . Yet transforming the first into the second costs $\frac{1}{2} \times 1 + \frac{1}{2} \times 1 = 1$. Shape matters beyond the mean.

4.1. Why squared distance needs a square root

Suppose one trip is twice as far. Ordinary distance makes it twice as expensive; squared distance makes it four times as expensive. The cost choice expresses what the comparison should penalize.

For any $p \geq 1$, use a genuine distance $d(x, y)$ between points, raise it to the power p , minimize, then take the p th root:

$$W_p(\mu, \nu) = \left(\min_{P \text{ feasible}} \sum_{i,j} P_{ij} d(x_i, y_j)^p \right)^{\frac{1}{p}}. \quad (13)$$

For $p = 2$, the root converts a mean squared travel distance into a root mean squared travel distance. For our four-unit example the squared cost of $P(t)$ is $55 - 18t$, so the same endpoint $t = 2$ minimizes it. After normalization,

$$W_2(\mu, \nu)^2 = \frac{2 \times 1^2 + 1 \times 4^2 + 1 \times 1^2}{4} = \frac{19}{4}, \quad W_2(\mu, \nu) = \frac{\sqrt{19}}{2} \approx 2.179. \quad (14)$$

Do not call $19/4$ the W_2 distance: it is its **square**. The root matters for the rules of distance. For point masses at $0, 1, 2$, squared travel costs would give 4 for 0 -to- 2 but only $1+1$ via 1 , violating the rule that a direct distance cannot exceed the sum of two intermediate distances. The rooted values give 2 and $1+1$ instead.

We take the general metric theorem as given: with a ground metric and finite p th moments, W_p is symmetric, is zero exactly for identical distributions, and satisfies the triangle inequality [1]. A finite p th moment means the average p th power of distance to a fixed point is finite. For finitely many points this is automatic. An arbitrary cost matrix does not automatically define a Wasserstein distance. Also, different plans can attain the same minimum; a unique distance value does not imply a unique plan.

5. Solving transport on a ruler

The line has an extra gift: we can order all the positions. Draw a vertical gate at coordinate r . Count how much mass is on its left before and after transport. If the starting side has too much, the excess must cross rightward; if it has too little, that deficit must cross leftward.

Call the mass at or to the left of r in μ the **cumulative distribution** $F_\mu(r)$. “Cumulative” just means we keep adding mass as the gate moves right. Define $F_\nu(r)$ the same way for the target. These are running totals, not the heights of the original piles.

At a gate between 1 and 3, the starting mass on the left is $3/4$; the target mass is $1/2$. Thus $1/4$ must cross to the right. In the optimal plan it is precisely the teal 0-to-4 route. No mass crosses back to the left.

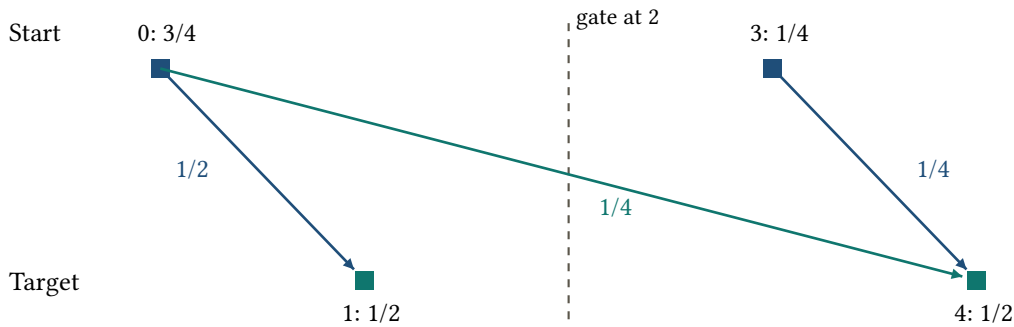


Figure 4: The dashed gate at 2 cuts the long teal route. Source mass on its left is $3/4$, target mass is $1/2$: exactly $1/4$ must leave this side. Line labels give route amounts, not distances.

At any gate, rightward crossing amount minus leftward crossing amount is $F_\mu(r) - F_\nu(r)$, fixed by the distributions. The **total** crossing amount is at least $|F_\mu(r) - F_\nu(r)|$. Sending material both ways wastes travel without helping the required net transfer.

5.1. Count each trip by the gates it crosses

A trip from 0 to 4 crosses every gate in an interval of width 4. Its contribution to total distance is its amount multiplied by that width. Instead of adding trip by trip, we can therefore add **gate by gate**: crossing amount $\times$ a small width, over the whole ruler. This counts exactly the same travel, simply in a different order.

The cumulative totals for our example have three nonzero gaps:

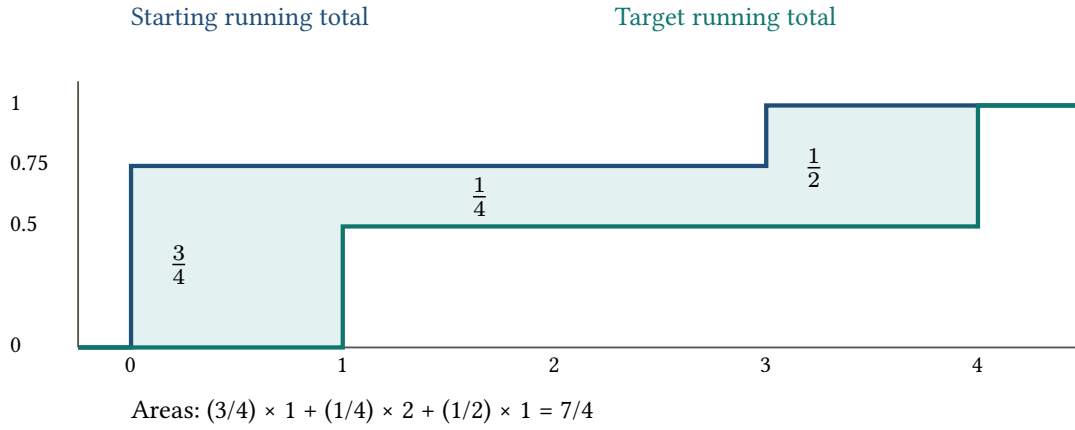


Figure 5: Navy is the starting cumulative total; teal is the target cumulative total. The shaded gap height is the necessary crossing amount. Multiply each height by its interval width to obtain area, hence transport cost.

This area gives a lower bound for any plan. To attain it, we need a plan that never sends mass in opposite directions across the same gate.

5.2. Build such a plan by consuming piles in order

Start with the leftmost source and leftmost unfilled target. Send the smaller of the remaining source amount and target demand. If the source empties, advance to the next source; if the target fills, advance to the next target. If both happen, advance both. Continue until all mass is assigned.

For the normalized example, send $\frac{1}{2}$ from 0 to 1. Target 1 fills; source 0 still has $\frac{1}{4}$. Send that remainder from 0 to 4, then send $\frac{1}{4}$ from 3 to 4. This is our original plan reconstructed by an algorithm.

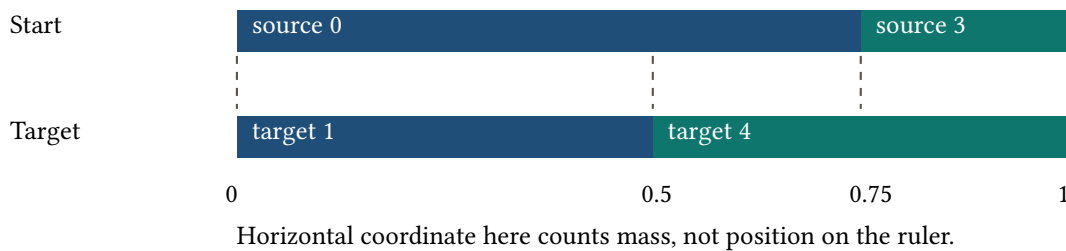


Figure 6: A common strip counts mass from 0 to 1. Source 0 occupies the first $\frac{3}{4}$; target 1 occupies only the first $\frac{1}{2}$. The overlap widths are the route amounts: $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{4}$. The middle slice must split away from the pile at 0.

Why does this construction attain the bound? It pairs earlier source mass with earlier target mass. Two opposed crossings at a gate would instead pair a left source with a right target and a right source with a left target, reversing that order. Our construction never does that. At every gate the crossing is entirely in the direction needed by the net deficit. Thus its crossing amount equals the absolute cumulative gap, and the lower bound is attained.

We have derived, for finite distributions on the real line with ordinary distance cost,

$$W_1(\mu, \nu) = \int_{-\infty}^{\infty} |F_\mu(r) - F_\nu(r)| \, dr. \quad (15)$$

The integral means the **total area** of the gap. In our finite example it is just the sum of three rectangles. The same formula extends to continuous distributions with finite first moments [1]. There one has smooth or partly smooth running totals, and integration replaces summing rectangle areas. This particular shortcut is for W_1 on a line, not a formula for squared cost or general multidimensional transport.

The ordered matching method also minimizes d^p on a line for $p \geq 1$ [1]. We have supplied the gate proof only for $p = 1$; the more general result uses convexity of the powered distance.

6. How a computer chooses the routes

For general finite point sets, use the same three ingredients: starting amounts, ending amounts, and pairwise costs. Coordinates can be two-dimensional positions, colors, or feature vectors, provided the cost expresses the comparison you intend. Changing that cost changes what “similar” means.

The objective in Equation 5 is linear: each route amount is multiplied by a fixed cost and the contributions are added. The constraints are also linear sums. This makes it a **linear program**. A transport or linear-programming solver can find the optimum to its numerical tolerance. The special one-dimensional construction is often simpler when its assumptions apply. A general solver must check all row and column totals, not just whether the total mass is right.

The numerical examples in this note are reproduced by

`scripts/verify_optimal_transport.py`, using Python’s standard library.

6.1. Optional extension: why add entropy?

This extension explains a common computational alternative. The core transport problem above is already complete. We will use one stated structural theorem to explain how this alternative is computed.

For probability mass, entropy measures how spread out the route probabilities are. Its useful sign convention here is

$$H(P) = - \sum_{i,j} P_{ij} (\ln P_{ij} - 1), \quad (16)$$

with $0 \ln 0$ interpreted as 0. The function $\ln$ is the natural logarithm, the inverse of the exponential e^x . The extra -1 changes entropy only by the fixed total mass; it does not change which plan wins. Splitting a probability into smaller positive entries increases this entropy.

Keep the same exact constraints, but minimize

$$\sum_{i,j} P_{ij} C_{ij} - \varepsilon H(P), \quad \varepsilon > 0. \quad (17)$$

The first term rewards short trips; the second rewards spreading the plan. The parameter ε chooses the tradeoff and has the same units as the per-unit cost. A positive value generally gives some mass to routes absent from the unregularized optimum. It does **not** relax conservation.

For finite costs and strictly positive source and target weights, the regularized problem has a unique positive solution of the form

$$P_{ij} = u_i K_{ij} v_j, \quad K_{ij} = e^{-\frac{C_{ij}}{\varepsilon}}. \quad (18)$$

We take this structure theorem, and the convergence of the scaling method below under these assumptions, from §4.2 of [1]. The derivation uses constrained optimization; it is not

assumed knowledge in this introduction. Zero-weight sources or destinations can first be removed.

The exponential converts cost into a positive **preference weight**: expensive routes get small weights. K alone is not a transport plan because its row and column totals usually differ from the required amounts. The positive factors u_i and v_j multiply each row and each column to fix those totals.

6.2. Derive the two scaling updates

Hold the column factors v_j fixed. The sum of row i in Equation 18 is $u_i \sum_j K_{ij} v_j$. To make it equal a_i , divide the wanted amount by the current sum before its row factor:

$$u_i \leftarrow \frac{a_i}{\sum_j K_{ij} v_j}. \quad (19)$$

Then hold these row factors fixed. Column j has sum $v_j \sum_i u_i K_{ij}$. Set it to b_j by the same reasoning:

$$v_j \leftarrow \frac{b_j}{\sum_i u_i K_{ij}}. \quad (20)$$

Initialize all $v_j = 1$, and alternate. These are **Sinkhorn's updates**. Fixing rows can disturb columns; fixing columns can disturb rows. Repeating is necessary until **both** sets of totals are sufficiently close to their targets.

To see the action with simple fractions, choose $\varepsilon = \frac{1}{\ln} 2$ for our distance matrix. Then $K_{ij} = 2^{-C_{ij}}$, so

$$K = \begin{pmatrix} \frac{1}{2} & \frac{1}{16} \\ \frac{1}{4} & \frac{1}{2} \end{pmatrix}. \quad (21)$$

Its row totals are $9/16$ and $3/4$, while we need $3/4$ and $1/4$. Multiply row one by $4/3$ and row two by $1/3$. The new grid is

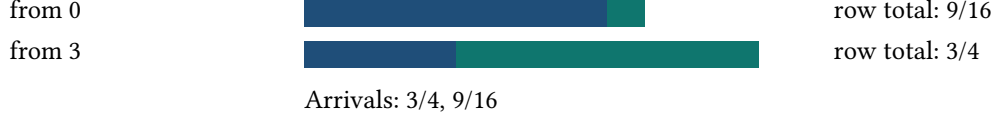
$$P^{\text{row}} = \begin{pmatrix} \frac{2}{3} & \frac{1}{12} \\ \frac{1}{12} & \frac{1}{6} \end{pmatrix}. \quad (22)$$

Now rows are right, but columns total $3/4$ and $1/4$. To obtain $1/2$ in each, multiply column one by $2/3$ and column two by 2 :

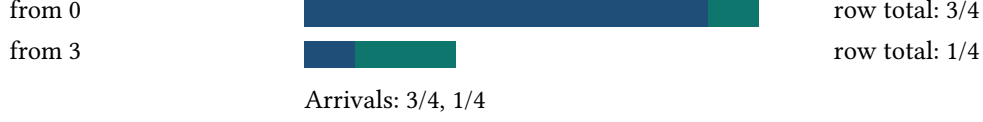
$$P^{\text{col}} = \begin{pmatrix} \frac{4}{9} & \frac{1}{6} \\ \frac{1}{18} & \frac{1}{3} \end{pmatrix}. \quad (23)$$

Columns are right; row totals are now $11/18$ and $7/18$. Neither of these intermediate grids is the finished answer.

A. Cost preferences K



B. Fix the starting piles



C. Fix the arrivals

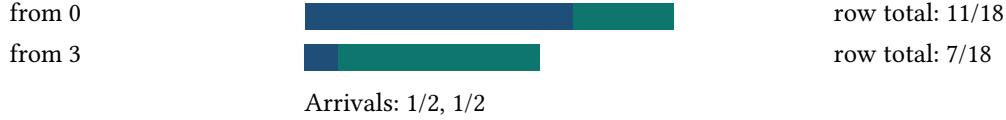


Figure 7: Bar width is route amount, on the same scale in every stage. Navy goes to destination 1; teal goes to destination 4. Row scaling stretches or shrinks a whole bar. Column scaling changes every segment of the same color. The printed totals reveal why one pass cannot satisfy both constraints.

At convergence, the script computes, for this ε ,

$$P_\varepsilon \approx \begin{pmatrix} 0.475569 & 0.274431 \\ 0.024431 & 0.225569 \end{pmatrix}. \quad (24)$$

Both row and column totals now match the required distributions. Its **travel cost** is approximately 1.847724, above the exact $W_1 = 1.75$. The extra leftward route from 3 to 1 is accepted in exchange for increased entropy. The full regularized objective also includes the entropy term, so it is a different number from this travel cost.

6.3. What the approximation guarantees, and what to check

In the finite problem, as ε tends to zero the regularized travel cost tends to the unregularized optimum. If several plans minimize that cost, regularization selects the one with maximal entropy in the limit [1]. Small ε also makes exponentials very small: naive arithmetic can underflow. Numerical implementations often work with logarithms of the factors instead; this note's moderate example does not need that.

After every full row/column update, check the largest absolute row error and the largest absolute column error. A correct column total alone is not a stopping rule. Moreover, a small marginal error certifies feasibility to tolerance, not unregularized optimality: ε still changes the objective. The regularized objective and the regularized plan's travel cost are not automatically Wasserstein distances.

The practical choice is now concrete: use ordered matching for the applicable line problem, a transport linear program for the original finite objective, or Sinkhorn for the explicitly regularized problem. In every case the same route grid connects supply, demand, geometry, and cost.

7. Sources and next steps

Gabriel Peyré and Marco Cuturi's *Computational Optimal Transport* [1] supports the terminology and formulations used here: Chapter 2 for distributions, plans and maps; §2.4 for Wasserstein's metric properties; §2.6 for ordered matching and cumulative distributions; and §§4.1–4.2 for entropy, scaling, convergence, and numerical stability. The numerical example, gate proof for finite W_1 , and explanatory figures are worked out in this note.

The central construction to retain is the route grid. Its entries say **how much**, its row and column totals say **what must be conserved**, and the cost grid says **which movements are expensive**. Minimizing those movements gives a way to compare entire distributions while respecting where their mass lives.

For an application after these foundations, the site's Motion Planning via Optimal Transport^o note uses a Sinkhorn step inside a robot motion-planning method. It assumes more optimization background than this introduction.

Bibliography

- [1] G. Peyré and M. Cuturi, “Computational Optimal Transport.” [Online]. Available: <https://arxiv.org/abs/1803.00567v4>^o